LEVODOPA + CARBIDOPA

Time-critical medicine

Never stop abruptly

CHEMICAL NAME Levodopa: (2S)-2-amino-3-(3,4-dihydroxyphenyl)propanoic acid + carbidopa: (2S)-3-(3,4-dihydroxyphenyl)-2-hydrazinyl-2-methylpropanoic acid

CLASS Dopamine precursor + peripheral dopa-decarboxylase inhibitor (co-careldopa)

TRADE NAMES Sinemet · Sinemet CR · Stalevo (with entacapone) · Duodopa (intestinal gel)

THERAPEUTIC USES / INDICATIONS

- Parkinson's disease; most effective symptomatic treatment.
- Restless legs syndrome (low dose).

ADMINISTRATION

- **PO** at precise times (time-critical — give within 30 min of prescribed time in hospital).
- Ideally 30–60 min before meals (high-protein meals ↓ absorption).
- NBM patients: do not omit — NG tube (dispersible) or alternative (rotigotine patch) via specialist.
- CR tablets not crushed.

DOSAGES

- **Start:** levodopa 50–100 mg three times daily (with carbidopa), increase gradually.
- **Usual:** 300–800 mg levodopa/day in 3–6 doses.
- Carbidopa ≥75 mg/day needed to prevent peripheral side effects.

PHARMACOKINETICS

- **Onset:** 30–60 min. **Duration:** 3–4 h (shortens with disease progression → wearing off).
- Absorbed by amino-acid transporter (competes with dietary protein).
- Carbidopa doesn't cross BBB. Levodopa **t½:** 1.5 h.

NURSING MANAGEMENT

- Give on time; never omit; escalate if unable to swallow.
- Monitor motor symptoms, dyskinesias, postural BP, falls.
- Monitor for hallucinations and confusion.
- Avoid dopamine-blocking antiemetics.

PHARMACODYNAMICS

- Levodopa crosses the BBB and is converted to **dopamine** in remaining nigrostriatal neurons.
- Carbidopa blocks **peripheral dopa decarboxylase** → more levodopa reaches brain; ↓ nausea and hypotension.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, anorexia.
- Postural hypotension, dizziness.
- Sleepiness; dark urine/sweat.

ADVERSE EFFECTS (SERIOUS)

- Dyskinesias (involuntary movements) and motor fluctuations (wearing-off, on–off).
- Hallucinations, psychosis, confusion; impulse-control disorders.
- Sudden sleep onset.
- Parkinsonism-hyperpyrexia syndrome (NMS-like) if stopped suddenly.

CONTRAINDICATIONS & CAUTIONS

- Angle-closure glaucoma.
- MAOI (non-selective) within 14 days.
- Melanoma or suspicious skin lesions (caution).
- Severe psychosis.

KEY INTERACTIONS

- **Antipsychotics (haloperidol), metoclopramide, prochlorperazine:** block dopamine — avoid (use domperidone for nausea).
- **Iron, high-protein meals:** ↓ absorption.
- **Non-selective MAOIs:** hypertensive crisis.

PATIENT EDUCATION

- Take at the same times every day; keep a symptom diary.
- Take 30–60 min before meals; spread protein through the day.
- Do not stop suddenly.
- Report hallucinations, compulsive behaviours, sudden sleepiness (avoid driving).

KEY POINTS

- Dopamine precursor + decarboxylase inhibitor.
- Dyskinesia, wearing-off, hallucinations.
- Time-critical — never omit.
- Avoid metoclopramide/haloperidol.